Precautions for LiDAR Mapping and Navigation

1. Experiment Objective

Understand the key precautions before LiDAR mapping and navigation, avoid common failure scenarios, and ensure that the mapping and navigation experiments proceed smoothly.

2. Experiment Principle

1. Explanation of Key Terms

Keyword	Description
LiDAR	A sensor that uses laser ranging to obtain distance data of the surrounding environment. This document uses the T-mini Plus 2D LiDAR, which outputs 360° scan data (`/scan` topic)
/scan topic	The standard topic name in ROS2 on which the LiDAR publishes scan data; it contains distance and angle information and is the core data source for mapping and navigation
T-mini Plus	The 2D LiDAR model used in this document; it supports 360° scanning with a measuring range of 0.03–12 m, and communicates with the Raspberry Pi via a USB serial port
SLAM	Simultaneous Localization and Mapping; building a map while determining its own position in an unknown environment
Nav2	The ROS2 navigation framework (Navigation2); it implements path planning, obstacle avoidance, and autonomous navigation based on an existing map

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑ Supported
No-PTZ version	☑ Supported
Required peripheral	T-mini Plus LiDAR

2. Pre-Mapping Checks

1. The LiDAR rotates normally, and `ros2 topic echo /scan` outputs data
2. The car is placed in the center of the environment, with at least 2 m of space around it
3. The ground is flat to reduce vibration
4. Unnecessary ROS2 nodes are shut down to save CPU

3. Pre-Navigation Checks

1. The map files (`.yaml` + `.pgm`) exist in the maps directory
2. The `/scan` topic has real-time data
3. The TF tree is complete (`odom` → `base_footprint` → `base_link` → `laser_Link`)
4. The initial pose matches the actual position (if not, manual relocation is required)

4. Environmental Requirements

- Avoid large areas of glass or mirrors (laser light will pass through / be reflected)
- Avoid empty halls (a lack of feature points is not conducive to scan matching)
- Avoid too many moving objects (people moving around will interfere with SLAM)

4. Experimental Phenomena

- During mapping, the map in RViz2 expands gradually
- During navigation, the robot can plan a path and avoid obstacles
- When the environmental requirements are met, the mapping quality and navigation stability improve significantly

5. Program Analysis

This section contains general precautions and does not involve any specific program. For the program analysis of each function, please refer to the corresponding chapters:

- Gmapping mapping: `6. Gmapping Mapping/gmapping-mapping.md`
- Cartographer mapping: `7. Cartographer Mapping/cartographer-mapping.md`
- slam_toolbox mapping: `9. slam_toolx Mapping/slam-toolx-mapping.md`
- Nav2 navigation: `10. Navigation2 Single-Point Navigation/navigation2-single-point-navigation.md`

6. Common Issues

Problem	Cause	Solution
The map does not update	Abnormal LiDAR data	Use `ros2 topic hz /scan` to check the frequency; normally about 6-8 Hz
Map distortion	The car is slipping or vibrating too much	Reduce the speed and choose flat ground
Navigation spins in circles	Incorrect initial pose	Use RViz 2D Pose Estimate to set it manually
The map fails to load	File path or permission issue	Confirm that the image path in `.yaml` is correct
Discontinuous LiDAR data	Unstable USB serial port or insufficient power supply	Check the LiDAR power supply and data cable connection
Localization drift	Sparse environmental features	Add more environmental features or switch to Cartographer (better loop closure detection)